

Database Design

AI-Based Smart Quiz System

Team Name: CSE4204-8A-T03

Database Name: ai_smart_quiz_db

Tables -

The database comprises four tables. Each table, its purpose, and its fields are described below.

Table: users

Purpose: Stores user registration and profile data for both teachers and students.

Field Name	Data Type	Constraints / Notes
id	INT	Primary Key, Auto Increment
name	VARCHAR	User's full name
email	VARCHAR	Unique, required
password	VARCHAR	Hashed password
is_teacher	BOOLEAN	1 = Teacher, 0 = Student
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp
email_verified_at	TIMESTAMP	Nullable; email verification timestamp
remember_token	VARCHAR	Nullable; session remember token

4.2 Table: courses

Purpose: Stores course information created and managed by teachers.

Field Name	Data Type	Constraints / Notes
id	INT	Primary Key, Auto Increment
teacher_id	INT	Foreign Key referencing users.id
name	VARCHAR	Course name
description	TEXT	Nullable; course description
time_per_question	INT	Default 30; seconds allowed per question
created_at	TIMESTAMP	Record creation timestamp

updated_at	TIMESTAMP	Record update timestamp
------------	-----------	-------------------------

4.3 Table: questions

Purpose: Stores multiple-choice questions associated with a specific course.

Field Name	Data Type	Constraints / Notes
id	INT	Primary Key, Auto Increment
course_id	INT	Foreign Key referencing courses.id
level	INT	Nullable; difficulty level
question	TEXT	Question text
option1	VARCHAR	Answer option 1
option2	VARCHAR	Answer option 2
option3	VARCHAR	Answer option 3
option4	VARCHAR	Answer option 4
correct_answer	VARCHAR	Correct answer value
explanation	TEXT	Explanation for the correct answer
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp

4.4 Table: quiz_attempts

Purpose: Records each student's quiz attempt, including performance metrics and anti-cheating indicators.

Field Name	Data Type	Constraints / Notes
id	INT	Primary Key, Auto Increment
user_id	INT	Foreign Key referencing users.id
course_id	INT	Foreign Key referencing courses.id
score	INT	Student's score for the attempt
total_questions	INT	Total number of questions in the quiz
time_taken	VARCHAR	Nullable; formatted time taken
seconds_taken	INT	Nullable; total seconds taken
is_suspicious	BOOLEAN	Default false; cheating flag
tab_switch_count	INT	Default 0; number of tab switches detected
avg_time_per_question	FLOAT	Nullable; average time per question

started_at	TIMESTAMP	Nullable; quiz start timestamp
created_at	TIMESTAMP	Record creation timestamp
updated_at	TIMESTAMP	Record update timestamp

5. Relationships

The following relationships exist between the tables in the ai_smart_quiz_db database.

- **users (Teacher) → courses** — One-to-Many. The teacher_id field in the courses table references the id field in the users table. A single teacher may create and manage multiple courses.
- **users (Student) → quiz_attempts** — One-to-Many. The user_id field in the quiz_attempts table references the id field in the users table. A single student may have multiple quiz attempt records across different courses.
- **courses → questions** — One-to-Many. The course_id field in the questions table references the id field in the courses table. A single course may contain many questions.
- **courses → quiz_attempts** — One-to-Many. The course_id field in the quiz_attempts table references the id field in the courses table. A single course may be associated with multiple quiz attempts from different students.

ER Diagram :

